Elasticsearch has come a long way in enabling organizations to utilize its powerful features in the search and analytics space. In addition to search and analytics use cases, it is used for application and infrastructure log analytics, enterprise security and threat detection, application performance and monitoring, distributed datastores, and more.

In this chapter, we examine the search space in general and skim over the evolution of search, from a traditional database-backed search to current modern search engines and their many convenient features. Along the way, we introduce Elasticsearch, the ultra-fast open source search engine, and examine its features, use cases, and customer adoption.

We also take a quick look at how generative artificial intelligence (AI) tools are beginning to unravel and disrupt the search space. With the advent of ChatGPT, a race to embrace AI and become the leader in the search space has begun. We dedicate a section to introducing the current players and exploring the future of AI-led search engines.

1.1 What makes a good search engine?

Let's take a moment to consider what makes a search engine "good" in terms of everyday experience. To help picture this, let's consider an experience I had with a bad search engine.



Recently my family adopted a puppy, Milly (that's her, pictured!). As we are first-time owners of a pup, we started searching online for dog food. I browsed my preferred grocer's website, but to my disappointment, the search results were not what I was looking for. The list of results included "poop bags" (the last thing you expect when searching for dog food) and other non-relevant products. The website also didn't have filters, dropdowns, or price-range selectors; it was just a plain page showing the search results, with clunky pagination enabled.

Not happy with my current grocer's search (curiosity overtook me—I was eager to find out how other search engines were implemented), I took my search to other grocers. One website presented a pet harness, while others were more hit and miss, including a search that showed me a baby's lunch box!

Not only did I get poor results from my search, but the search engines behind these supermarkets also didn't provide suggestions while I typed; nor did they correct my input when I misspelled *dog food* as *dig fod* (we are spoiled by Google when it comes to suggestions and corrections—we expect every search engine to have features similar to Google's). Most didn't bother suggesting alternatives or similar items. Some of the results weren't relevant (i.e., based on relevancy—although this can be forgiven, as not all search engines are expected to produce relevant results). One grocer returned 2,400 results for a simple search!